

DIGITAL FORENSICS

Supplementary Materials Provided:

2 Log Files

- i) pre_auth.log
- ii) post_auth.log

Part A: Log Analysis & Timeline Reconstruction (60 marks)

1) Identify the attacker's IP address (2 marks)

- Justify with evidence from the logs.

2) Find the exact time of successful unauthorized access (3 marks)

- Include source IP, username, and timestamp.

3i) List all commands executed by the attacker (5 marks)

- Include timestamps.

4) What malicious file was downloaded? From where? (4 marks)

5) Compare pre-attack and post-attack logs. List three suspicious differences (6 marks)

- Example: new brute-force patterns, new usernames, etc.

6) Reconstruct the full attack timeline in this format (20 marks):

Time	Event	Evidence (log line reference)
...	...	...

Minimum 8 events required.

7) Which legitimate user accounts were likely compromised? (4 marks)

- Explain your reasoning.

8) Post-attack, what persistence mechanism might the attacker have installed? (4 marks)

- From the logs, what suggests this?

9) Write a one-paragraph executive summary of the incident (12 marks)

- Target audience: CISO (non-technical but security-aware).

Part B: Theoretical & Forensic Questions (40 marks)

Chain of custody – Explain why you cannot simply edit these log files directly. What steps would you take before analysis? (8 marks)

Log integrity – How could the attacker erase their tracks from auth.log? How would a forensic investigator detect such tampering? (8 marks)

Timeline tools – Name two CLI tools (Linux) to merge, sort, and analyze these logs efficiently. Show an example command. (6 marks)

False positives – In the pre-attack log, there are failed logins from 203.0.113.5. Are they part of the attack? Why or why not? (6 marks)

Legal & ethical – If this was a real incident, would you be allowed to examine Alice's sudo commands without her consent? Under what framework? (6 marks)

Reporting – List 5 essential sections of a digital forensics investigation report. (6 marks)